

JEE–MAIN EXAMINATION – JANUARY 2025

(HELD ON THURSDAY 23rd JANUARY 2025)

TIME : 9 : 00 AM TO 12 : 00 NOON

PHYSICS

TEST PAPER WITH SOLUTION

SECTION-A

26. Regarding self-inductance :

A : The self-inductance of the coil depends on its geometry.

B : Self-inductance does not depend on the permeability of the medium.

C : Self-induced e.m.f. opposes any change in the current in a circuit.

D : Self-inductance is electromagnetic analogue of mass in mechanics.

E : Work needs to be done against self-induced e.m.f. in establishing the current.

Choose the correct answer from the options given below:

(1) A, B, C, D only (2) A, C, D, E only

(3) A, B, C, E only (4) B, C, D, E only

Ans. (2)

Sol. Self inductance of coil

$$L = \frac{\mu_0 N^2 A}{2\pi R}$$

27. A light hollow cube of side length 10 cm and mass 10g, is floating in water. It is pushed down and released to execute simple harmonic oscillations.

The time period of oscillations is $y\pi \times 10^{-2}$ s, where the value of y is

(Acceleration due to gravity, $g = 10 \text{ m/s}^2$, density of water = 10^3 kg/m^3)

(1) 2 (2) 6

(3) 4 (4) 1

Ans. (1)

Sol. $a^2 \times \rho g = m a_{\text{net}}$

$$\frac{L^2 \rho g}{m} x = a_{\text{net}}$$

$$T = 2\pi \sqrt{\frac{m}{L^2 \rho g}}$$

where $m = 10\text{g}$, $L = 10 \text{ cm}$, $\rho = 1000 \text{ kg/m}^3$

28. Given below are two statements:

Statement-I : The hot water flows faster than cold water.

Statement-II : Soap water has higher surface tension as compared to fresh water.

In the light above statements, choose the **correct** answer from the options given below

(1) Statement-I is false but Statement II is true

(2) Statement-I is true but Statement II is false

(3) Both Statement-I and Statement-II are true

(4) Both Statement-I and Statement-II are false

Ans. (2)

Sol. Hot water is less viscous than cold water.

Surfactant reduces surface tension.

29. A sub-atomic particle of mass 10^{-30} kg is moving with a velocity $2.21 \times 10^6 \text{ m/s}$. Under the matter wave consideration, the particle will behave closely like _____. ($h = 6.63 \times 10^{-34} \text{ J.s}$)

(1) Infra-red radiation (2) X-rays

(3) Gamma rays (4) Visible radiation

Ans. (2)

$$\text{Sol. } \lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{10^{-30} \times 2.21 \times 10^6} = 3 \times 10^{-10} \text{ m}$$

Hence particle will behave as x-ray.

30. A spherical surface of radius of curvature R, separates air from glass (refractive index = 1.5). The centre of curvature is in the glass medium. A point object 'O' placed in air on the optic axis of the surface, so that its real image is formed at 'I' inside glass. The line OI intersects the spherical surface at P and $PO = PI$. The distance PO equals to-

(1) 5R

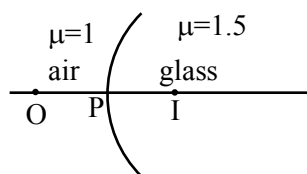
(2) 3R

(3) 2R

(4) 1.5R

Ans. (1)

Sol.



$$PO = u = -x$$

$$PI = v = x$$

$$PO = PI$$

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{x} + \frac{1}{x} = \frac{1}{2R}$$

$$\frac{5}{2x} = \frac{1}{2R}$$

$$X = 5R$$

31. A radioactive nucleus n_2 has 3 times the decay constant as compared to the decay constant of another radioactive nucleus n_1 . If initial number of both nuclei are the same, what is the ratio of number of nuclei of n_2 to the number of nuclei of n_1 , after one half-life of n_1 ?

- (1) $1/4$ (2) $1/8$
(3) 4 (4) 8

Ans. (1)

Sol. $N_2 = N_0 e^{-\lambda_2 t}$

$$N_1 = N_0 e^{-\lambda_1 t}$$

$$\frac{N_2}{N_1} = e^{-2\lambda_1 t}$$

$$t_{\text{half life of } N_1} t = \frac{\ln 2}{\lambda_1} n_1$$

$$\frac{N_0}{2} = N_0 e^{-\lambda_1 t}$$

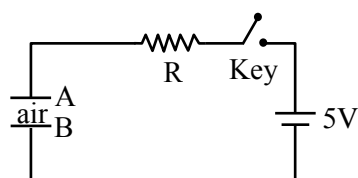
$$\lambda_1 t = \ln 2$$

$$t = \frac{\ln 2}{\lambda_1}$$

$$= e^{-2\lambda_1 \frac{\ln 2}{\lambda_1}}$$

$$\frac{N_2}{N_1} = \frac{1}{4}$$

32. Identify the valid statements relevant to the given circuit at the instant when the key is closed.



- A. There will be no current through resistor R.
B. There will be maximum current in the connecting wires.
C. Potential difference between the capacitor plates A and B is minimum.
D. Charge on the capacitor plates is minimum.

Choose the correct answer from the options given below :

- (1) C, D only (2) B, C, D only
(3) A, C only (4) A, B, D only

Ans. (2)

Sol. Initially capacitor behave as a short circuit.

So current will be maximum.

Charge on capacitor will be zero.

Potential difference across capacitor will be zero.

33. The position of a particle moving on x-axis is given by $x(t) = A \sin t + B \cos^2 t + Ct^2 + D$, where t

is time. The dimension of $\frac{ABC}{D}$ is-

- (1) L (2) $L^3 T^{-2}$
(3) $L^2 T^{-2}$ (4) L^2

Ans. (3)

Sol. Dimension $[x(t)] = [L]$

$$[A] = [L]$$

$$[B] = [L]$$

$$[C] = [LT^{-2}]$$

$$[D] = [L]$$

$$\left[\frac{ABC}{D} \right] = \left[\frac{L \times L \times LT^{-2}}{L} \right] = [L^2 T^{-2}]$$

34. Match the List-I with List-II

List-I		List-II	
A.	Pressure varies inversely with volume of an ideal gas.	I.	Adiabatic process
B.	Heat absorbed goes partly to increase internal energy and partly to do work.	II.	Isochoric process
C.	Heat is neither absorbed nor released by a system	III.	Isothermal process
D.	No work is done on or by a gas	IV.	Isobaric process

Choose the **correct** answer from the options given below :

- (1) A-I, B-IV, C-II, D-III
 (2) A-III, B-I, C-IV, D-II
 (3) A-I, B-III, C-II, D-IV
 (4) A-III, B-IV, C-I, D-II

Ans. (4)

Sol. $A \rightarrow P \propto \frac{1}{V}$

$\Rightarrow PV = \text{constant}$

$\Rightarrow nRT = \text{const.} \Rightarrow T = \text{const.}$

Hence Isothermal III

$B \rightarrow IV$

$W \neq 0, \Delta U \neq 0, \Delta Q \neq 0$ [only isobaric]

$C \rightarrow I \Delta Q = 0$ Adiabatic

$D \rightarrow II w = 0$ Isochoric

III IV I II

35. Consider a moving coil galvanometer (MCG) :

A : The torsional constant in moving coil galvanometer has dimensions $[ML^2T^{-2}]$

B : Increasing the current sensitivity may not necessarily increase the voltage sensitivity.

C : If we increase number of turns (N) to its double (2N), then the voltage sensitivity doubles.

D : MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.

E : Current sensitivity of MCG depends inversely on number of turns of coil.

Choose the correct answer from the options given below :

- (1) A, B only (2) A, D, only
 (3) B, D, E only (4) A, B, E only

Ans. (1)

Sol. (A) $\tau = C\theta \Rightarrow [ML^2T^{-2}] = [C][1]$

(B) $C.S. = \frac{\theta}{I} = \frac{BNA}{C}$;

$V.S. = \frac{BNA}{RC}$ [R also depends on 'N']

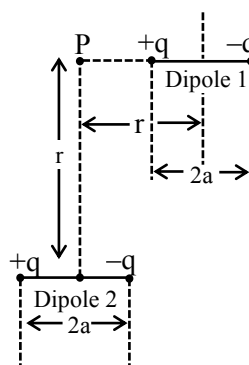
(C) $V.S. \propto \frac{NAB}{CR}$ $R \rightarrow NR$

(D) False [Theory]

(E) E [False] $C.S. \propto N$

$\Rightarrow \therefore C.S. = \frac{NAB}{C}$

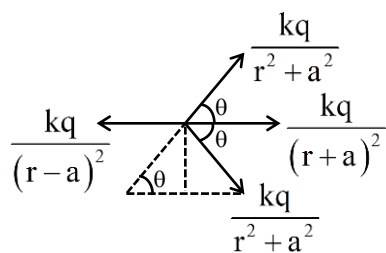
36. A point particle of charge Q is located at P along the axis of an electric dipole 1 at a distance r as shown in the figure. The point P is also on the equatorial plane of a second electric dipole 2 at a distance r. The dipoles are made of opposite charge q separated by a distance 2a. For the charge particle at P not to experience any net force, which of the following correctly describes the situation?



- (1) $\frac{a}{r} = 20$ (2) $\frac{a}{r} \sim 10$
 (3) $\frac{a}{r} \sim 0.5$ (4) $\frac{a}{r} \sim 3$

Ans. (4)

Sol.



$$\frac{kq}{(r-a)^2} = \frac{kq}{(r+a)^2} + \frac{2kq}{(r^2+a^2)} \cos \theta$$

$$\frac{1}{(r-a)^2} = \frac{1}{(r+a)^2} + \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} = \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\frac{4ra}{(r^2-a^2)^2} = \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{2r}{(r^2-a^2)^2} = \frac{1}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\frac{4r^2}{(r^2-a^2)^4} = \frac{1}{(r^2+a^2)^3}$$

$$\Rightarrow 4r^2(r^2+a^2)^3 = (r^2-a^2)^4$$

$$4r^8 \left(1 + \frac{a^2}{r^2}\right)^3 = r^8 \left(1 - \frac{a^2}{r^2}\right)^4$$

$$4 \left(1 + \frac{a^2}{r^2}\right)^3 = \left(1 - \frac{a^2}{r^2}\right)^4$$

Exact value cannot be solved in exam for this equation to be true

$$\left|\frac{a}{r}\right| > 1 \Rightarrow a > r$$

But point charge Q lies between charges of dipole 1 hence electric field cannot be zero.

There for it should be bonus.

But by solving from mathematical software we are getting $a/r \approx 3$.

37. A gun fires a lead bullet of temperature 300K into a wooden block. The bullet having melting temperature of 600 K penetrates into the block and melts down. If the total heat required for the process is 625 J, then the mass of the bullet is ____ grams.

(Latent heat of fusion of lead = $2.5 \times 10^4 \text{ JKg}^{-1}$ and specific heat capacity of lead = $125 \text{ JKg}^{-1} \text{ K}^{-1}$)

- (1) 20 (2) 15
(3) 10 (4) 5

Ans. (3)

Sol. $625 = ms\Delta T + mL$

$$625 = m[125 \times 300 + 2.5 \times 10^4]$$

$$625 = m[37500 + 25000]$$

$$625 = m[62500]$$

$$m = \frac{1}{100} \text{ kg}$$

$$M = 10 \text{ grams}$$

38. What is the lateral shift of a ray refracted through a parallel-sided glass slab of thickness 'h' in terms of the angle of incidence 'i' and angle of refraction 'r', if the glass slab is placed in air medium ?

- (1) $\frac{h \tan(i-r)}{\tan r}$ (2) $\frac{h \cos(i-r)}{\sin r}$
(3) h (4) $\frac{h \sin(i-r)}{\cos r}$

Ans. (4)

Sol. Formula base

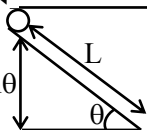
$$\frac{h \sin(i-r)}{\cos r}$$

39. A solid sphere of mass 'm' and radius 'r' is allowed to roll without slipping from the highest point of an inclined plane of length 'L' and makes an angle 30° with the horizontal. The speed of the particle at the bottom of the plane is v_1 . If the angle of inclination is increased to 45° while keeping L constant. Then the new speed of the sphere at the bottom of the plane is v_2 . The ratio of $v_1^2 : v_2^2$ is

- (1) $1 : \sqrt{2}$ (2) $1 : 3$
(3) $1 : 2$ (4) $1 : \sqrt{3}$

Ans. (1)

Sol. $L \sin \theta$



using WET

$$W_g = k_f - k_i$$

$$Mg L \sin \theta = k_f - k_i$$

$$\text{K.E. in pure rolling } \frac{1}{2} m V_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

$$= \frac{1}{2} m V^2 + \frac{1}{2} \times \frac{2}{5} m R^2 \frac{V^2}{R^2}$$

$$\frac{7}{10} m V^2$$

$$mgL \sin \theta = \frac{7}{10} m V_f^2 - 0$$

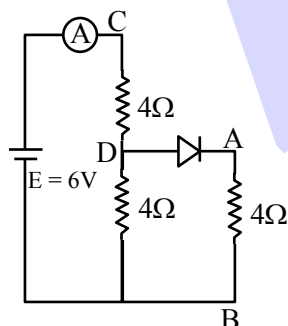
$$V_f^2 \propto \sin \theta$$

$$\left(\frac{V_1}{V_2} \right)^2 = \frac{\sin \theta_1}{\sin \theta_2} = \frac{\sin 30^\circ}{\sin 45^\circ} = \frac{1}{\sqrt{2}}$$

40. Refer to the circuit diagram given in the figure, which of the following observation are correct?

- A. Total resistance of circuit is 6Ω .
- B. Current in Ammeter is $1A$
- C. Potential across AB is 4 Volts.
- D. Potential across CD is 4 Volts.
- E. Total resistance of the circuit is 8Ω .

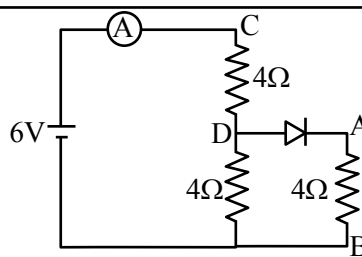
Choose the correct answer from the options given below:



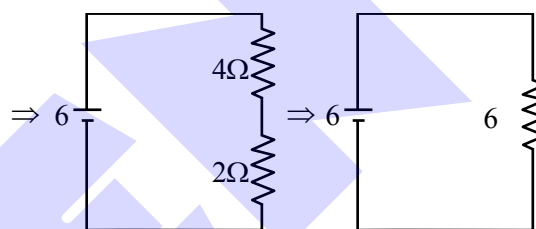
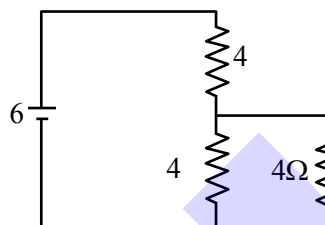
- (1) A, B and D only
- (2) A, C and D only
- (3) B, C and E only
- (4) A, B and C only

Ans. (1)

Sol.

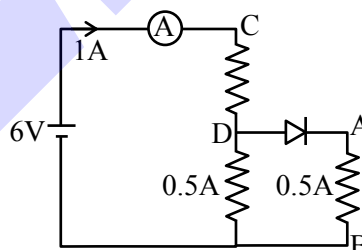


$\Downarrow$



Current through ammeter = $1 A$

$$R_{net} = 6\Omega$$



$$V_{AB} = 0.5 \times 4 = 2 \text{ volt}$$

$$V_{CD} = 1 \times 4 = 4 \text{ volt}$$

A, B & D are correct

41. The electric flux is $\phi = \alpha \sigma + \beta \lambda$

where λ and σ are linear and surface charge

density, respectively, $\left(\frac{\alpha}{\beta} \right)$ represents

- (1) charge
- (2) electric field
- (3) displacement
- (4) area

Ans. (3)

Sol. $\phi = \alpha\sigma + \beta\lambda$

$$[\phi] = [\alpha\sigma] = [\beta\lambda]$$

$$[\alpha] = \frac{[\phi]}{[\sigma]} \quad \left[\frac{\alpha}{\beta} \right] = \frac{[\lambda]}{[\sigma]}$$

$$[\beta] = \frac{[\phi]}{[\lambda]} = \frac{[Q/L]}{[Q/\text{Area}]} = \left[\frac{\text{Area}}{\text{Length}} \right]$$

$$\left[\frac{\alpha}{\beta} \right] = L$$

42. Given a thin convex lens (refractive index μ_2), kept in a liquid (refractive index μ_1 , $\mu_1 < \mu_2$) having radii of curvature $|R_1|$ and $|R_2|$. Its second surface is silver polished. Where should an object be placed on the optic axis so that a real and inverted image is formed at the same place ?

$$(1) \frac{\mu_1 |R_1| \cdot |R_2|}{\mu_2 (|R_1| + |R_2|) - \mu_1 |R_1|}$$

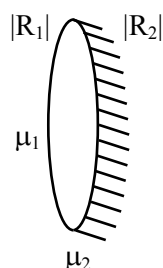
$$(2) \frac{\mu_1 |R_1| \cdot |R_2|}{\mu_2 (|R_1| + |R_2|) - \mu_1 |R_2|}$$

$$(3) \frac{\mu_1 |R_1| \cdot |R_2|}{\mu_2 (2|R_1| + |R_2|) - \mu_1 \sqrt{|R_1| \cdot |R_2|}}$$

$$(4) \frac{(\mu_2 + \mu_1) |R_1|}{(\mu_2 - \mu_1)}$$

Ans. (2)

Sol.



$$\frac{1}{f_{eq}} = \frac{2}{f_L} - \frac{1}{f_m}$$

$$f_m = -\frac{|R_2|}{2}$$

$$\frac{1}{f_L} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\frac{1}{f_{eq}} = 2 \left(\frac{\mu_2 - \mu_1}{\mu_1} \right) \left(\frac{R_1 + R_2}{R_1 R_2} \right) + \frac{2}{R_2}$$

$$= \frac{2}{R_2} \left[\frac{(\mu_2 - \mu_1)(R_1 + R_2) + \mu_1 R_1}{\mu_1 R_1} \right]$$

$$= \frac{2}{R_2} \left[\frac{\mu_2 R_1 + \mu_2 R_2 - \mu_1 R_1 - \mu_1 R_2 + \mu_1 R_1}{\mu_1 R_1} \right]$$

$$\frac{1}{f_{eq}} = \frac{2[\mu_2 R_1 + \mu_2 R_2 - \mu_1 R_2]}{\mu_1 R_1 R_2}$$

For same size of image

$$u = 2f$$

$$u = \frac{\mu_1 R_1 R_2}{\mu_2 R_1 + \mu_2 R_2 - \mu_1 R_2}$$

43. The electric field of an electromagnetic wave in free space is

$$\vec{E} = 57 \cos[7.5 \times 10^6 t - 5 \times 10^{-3} (3x + 4y)] (4\hat{i} - 3\hat{j}) \text{ N/C.}$$

The associated magnetic field in Tesla is-

$$(1) \vec{B} = \frac{57}{3 \times 10^8} \cos [7.5 \times 10^6 t - 5 \times 10^{-3} (3x + 4y)] (5\hat{k})$$

$$(2) \vec{B} = \frac{57}{3 \times 10^8} \cos [7.5 \times 10^6 t - 5 \times 10^{-3} (3x + 4y)] (\hat{k})$$

$$(3) \vec{B} = -\frac{57}{3 \times 10^8} \cos [7.5 \times 10^6 t - 5 \times 10^{-3} (3x + 4y)] (5\hat{k})$$

$$(4) \vec{B} = -\frac{57}{3 \times 10^8} \cos [7.5 \times 10^6 t - 5 \times 10^{-3} (3x + 4y)]$$

$$(\hat{k})$$

Ans. (3)

Sol. $\vec{K} = 3\hat{i} + 4\hat{j}$

$$\hat{K} = \frac{3\hat{i} + 4\hat{j}}{5}$$

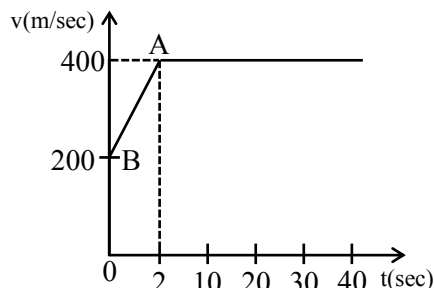
$$\hat{E} = \frac{4\hat{i} - 3\hat{j}}{5}$$

$$\hat{B} = \hat{K} \times \hat{E}$$

$$\hat{B} = -\hat{Z}$$

$$B_0 = \frac{E_0}{C} = \frac{57}{3 \times 10^8}$$

44. The motion of an airplane is represented by velocity-time graph as shown below. The distance covered by airplane in the first 30.5 second is _____ km.



- (1) 9 (2) 6
(3) 3 (4) 12

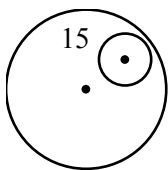
Ans. (4)

Sol. Total Area under curve.

45. Consider a circular disc of radius 20 cm with centre located at the origin. A circular hole of a radius 5 cm is cut from this disc in such a way that the edge of the hole touches the edge of the disc. The distance of centre of mass of residual or remaining disc from the origin will be-

- (1) 2.0 cm (2) 0.5 cm
(3) 1.5 cm (4) 1.0 cm

Ans. (4)



Sol.

mass of disc = m

mass of cut part = $\frac{m}{16}$

$$X_{\text{com}} = \frac{m \times 0 - \frac{m}{16} \times 15}{m - \frac{m}{16}}$$

= 1 cm.

46. A positive ion A and a negative ion B has charges $6.67 \times 10^{-19} \text{ C}$ and $9.6 \times 10^{-10} \text{ C}$, and masses $19.2 \times 10^{-27} \text{ kg}$ and $9 \times 10^{-27} \text{ kg}$ respectively. At an instant, the ions are separated by a certain distance r . At that instant the ratio of the magnitudes of electrostatic force to gravitational force is $P \times 10^{-13}$, where the value of P is _____.

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-1}$ and universal

gravitational constant as $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$)

Ans. (BONUS)

Sol.
$$\frac{9 \times 10^9 \times 6.67 \times 10^{-19} \times 9.6 \times 10^{-10}}{6.67 \times 10^{-11} \times 19.2 \times 10^{-27} \times 9 \times 10^{-27}}$$

$$\frac{1}{2} \times 10^{45}$$

Charge is not integral multiple of electron.

47. Two particles are located at equal distance from origin. The position vectors of those are represented by $\vec{A} = 2\hat{i} + 3n\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 4p\hat{k}$, respectively. If both the vectors are at right angle to each other, the value of n^{-1} is _____.

Ans. (3)

Sol. $\vec{A} \cdot \vec{B} = 0$

$$4 - 6n + 8p = 0$$

$$|\vec{A}| = |\vec{B}|$$

$$4 + 9n^2 + 4 = 4 + 4 + 16p^2$$

$$9n^2 = 16p^2$$

$$P = +\frac{3}{4}n$$

$$4 - 6n \pm 6n = 0$$

$$12n = 4$$

$$n = \frac{1}{3}$$

48. An ideal gas initially at 0°C temperature, is compressed suddenly to one fourth of its volume. If the ratio of specific heat at constant pressure to that at constant volume is $3/2$, the change in temperature due to the thermodynamics process is ____ K.

Ans. (273)

Sol. $\gamma = \frac{3}{2}$

$$TV^{\gamma-1} = C$$

$$273 V_0^{0.5} = T \left(\frac{V_0}{4} \right)^{0.5}$$

$$T = 273 \times 2 = 546$$

$$\Delta T = 273$$

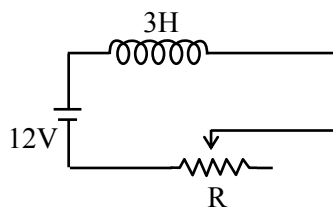
49. A force $\vec{f} = x^2y\hat{i} + y^2\hat{j}$ acts on a particle in a plane $x + y = 10$. The work done by this force during a displacement from $(0, 0)$ to $(4\text{m}, 2\text{m})$ is ____ Joule (round off to the nearest integer)

Ans. (152)

Sol. $\int_0^4 x^2(10-x)dx + \int_0^2 y^2 dy$

$$= \left[\frac{10x^3}{3} - \frac{x^4}{4} \right]_0^4 + \left[\frac{y^3}{3} \right]_0^2 = \frac{640}{3} - 64 + \frac{8}{3} = 152$$

50.



In the given circuit the sliding contact is pulled outwards such that electric current in the circuit changes at the rate of 8 A/s . At an instant when R is 12Ω , the value of the current in the circuit will be ____ A.

Ans. (3)

Sol. $\varepsilon - \frac{LdI}{dt} - IR = 0$

$$12 - 3 \times (-8) - I \times 12 = 0$$

$$I = 3$$

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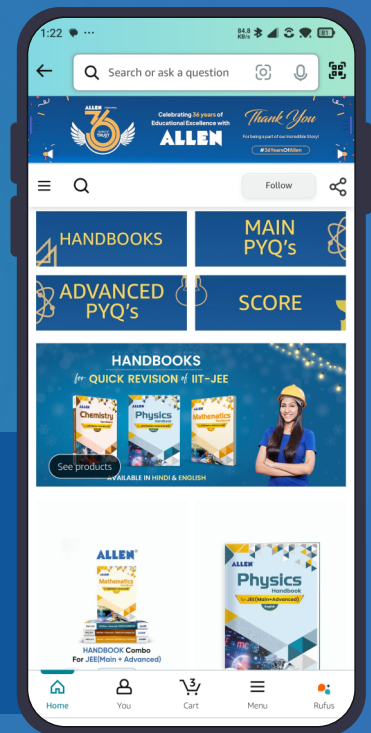


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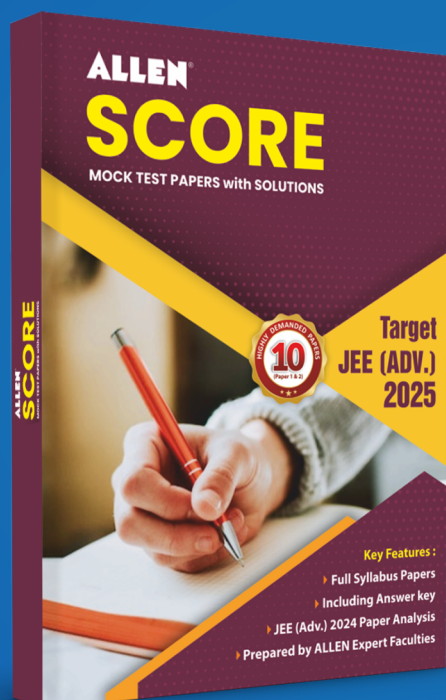


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